PROJECT LAB · FACTOR HUNTING

AI Factor Lens

Progress update: what changed this week, what the model shows, and what is still open

4 November 2026 · Based on commits a3ede42 → de64c6c

WHAT CHANGED

The notebooks became a reusable pipeline

01

Package refactor

Logic moved into src/factor_lens/, one module per stage. The four notebooks are now short entry points.

02

Versioned stages

Raw → clean → factors → models, each saved with a version tag, so any stage reruns without re-downloading.

03

Results committed

Clean data, factor library, model fits, attribution tables and figures are all in the repo.

4 commits by Santa David, 28 Oct – 3 Nov · 822 files changed

DATA · PHASES 1–2

Data is downloaded, cleaned and checked

621/623
tickers downloaded (2 came back empty)

2016–26
daily prices to 1 Oct 2026, with dividends and splits

~5,100
issues flagged, not deleted: 3.0k outliers, 1.4k stale prices

6/6
sanity checks pass (fund vs the index it tracks)

Sanity check	Correlation	Result
SPY vs ^GSPC	0.998	Pass
QQQ vs ^NDX	0.999	Pass
IWM vs ^RUT	0.996	Pass
SMH vs ^SOX	0.991	Pass
Equal-weight S&P stocks vs RSP	0.992	Pass

BASKET AND FACTORS · PHASES 3–5

39 AI-chain stocks, 12 factors to explain them

The basket: 7 layers

Layer	Examples
Chips	NVDA, AMD, AVGO, TSM, MU
Networking, servers	ANET, SMCI, DELL, CLS
Hyperscalers	MSFT, GOOGL, AMZN, META
Data centers	EQIX, DLR, CRWV, NBIS
Power generation	VST, CEG, NRG, TLN
Grid, electrical	ETN, VRT, PWR, GEV
Nuclear, uranium	CCJ, BWXT, LEU, OKLO

38 candidates → 12 in the model

Market, macro and style

MKT · D10Y · USD · OIL · LOWVOL

AI-adjacent

POWER · GRID · URANIUM

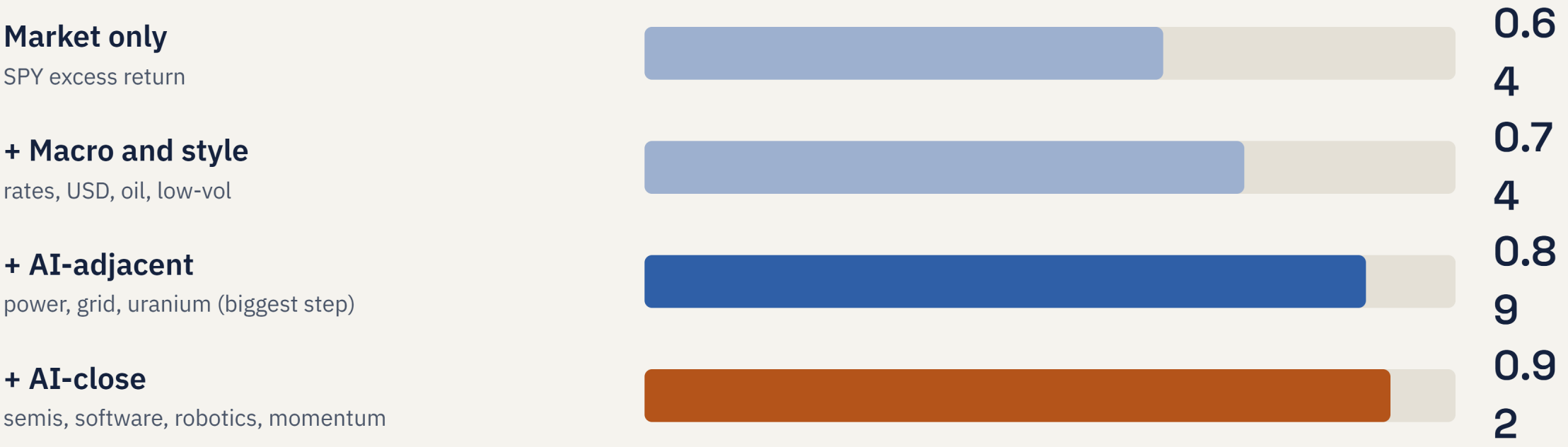
AI-close

SEMI · SOFT · ROBO · MOM

Picked by correlation clustering and VIF. Condition number falls from 354 to about 20 (max VIF 4.5). Orthogonalized with Gram–Schmidt.

MODEL RESULTS · PHASE 6

The factors explain 92% of the core basket out of sample



R^2 on the 2024-01 to 2026-10 test period, which the model never saw. Rolling 252-day regression, estimated on 2017–2023.

MODEL RESULTS BY LAYER

Chips follow AI factors; power and grid follow power factors

Layer (test-period R^2)	Market only	+ Macro, style	+ AI-adjacent	+ AI-close
Chips	0.57	0.70	0.76	0.91 (+0.15)
Networking, servers	0.45	0.55	0.65	0.71
Hyperscalers	0.61	0.65	0.67	0.74
Data centers	0.27	0.31	0.40	0.40
Power generation	0.26	0.30	0.83 (+0.53)	0.84
Grid, electrical	0.41	0.47	0.76 (+0.29)	0.77
Nuclear, uranium	0.24	0.27	0.68 (+0.41)	0.68
Core basket	0.64	0.74	0.89	0.92

Data centers and neoclouds stay poorly explained (0.40): short histories and stock-specific stories.

OPEN ISSUE · PLACEBO TEST

The AI factors help non-AI stocks more than AI stocks

+0.04

R^2 gain for AI stocks when the AI-close factors are added

+0.14

R^2 gain for non-AI controls: KO, PG, WMT, JNJ, XOM, JPM, MCD, CAT

If the factors were AI-specific, controls should gain little. One guess, not yet tested: momentum (MOM) acts as a general style factor.

STATUS AND NEXT STEPS

Data is done; methods and the report are next

Phase	Status	Still missing
0 · Design decisions	Partial	Written decision log, research question
1 · Data acquisition	Done	Tick off the GitHub issue
2 · Cleaning and QA	Done	Tick off the GitHub issue
3 · EDA and basket	Mostly	Cap-weighted basket variant
4 · Factor library	Mostly	Check against the issue checklist
5 · Selection, orthogonalization	Partial	LASSO, stability selection, PCA, Löwdin, comparison
6 · Regression	Mostly	EWMA betas; fix the placebo result
7 · Risk and P&L attribution	Partial	Euler risk split, waterfall report
8 · Robustness	Partial	Window, frequency, order, EW vs CW sweep
9 · Deliverables	Not started	Report, data dictionary, final notebooks